

00 · Foundations — the words the cookbook assumes

July 13, 2026

The notebook that comes **before** everything else. It builds the core vocabulary — potential outcomes, counterfactuals, confounding, selection bias, randomization, identification, the do-operator, DAGs, ATE vs ATT vs CATE — from nothing, with one marketing example, and works through an **estimator ladder**: naive $\rightarrow$ regression-adjust $\rightarrow$ IPW $\rightarrow$ doubly-robust — the naive rung fails on purpose, and each adjustment rung then recovers a known effect from confounded data. It then shows what **no rung can fix** (an unmeasured confounder, broken overlap), runs the full **Bayesian workflow** — prior $\rightarrow$ likelihood $\rightarrow$ posterior $\rightarrow$ R-hat/ESS — on one small model, replays the lesson on the most famous real dataset in causal inference, and closes by **pricing the identification error in euros** and mapping every later notebook to its estimand.

No new theory here — it mirrors `docs/causal_inference_primer.pdf`. Skim it once and every later notebook reads as a *named tool* rather than new vocabulary.

1 · Treatment, unit, outcome — and the counterfactual

A **treatment** is any action whose effect we want (send a discount, show an ad, change a price). The **unit** receives it (customer, store, region); the **outcome** is what we measure after.

For binary $T \in \{0, 1\}$ each unit has **two potential outcomes**: $Y(1)$ if treated, $Y(0)$ if not. We only observe **one** — $Y = TY(1) + (1 - T)Y(0)$ — so the individual effect $Y(1) - Y(0)$ is never seen. This **fundamental problem of causal inference** is a *missing-data* problem; everything downstream is principled imputation of the half of the table we didn't get to see.

	customer	Y(0) not emailed	Y(1) emailed	individual effect	emailed?	\
0	Anna	30	80	50	yes	
1	Ben	55	60	5	no	
2	Cara	20	22	2	yes	
	-> OBSERVED -> MISSING (counterfactual)					
0		80		30		
1		55		60		
2		22		20		

The trap in plain sight: comparing the **emailed** customers' *observed* spend to the **not-emailed** ones' is **not** the true average effect — that naive observed contrast is **confounded** by *who got chosen* to be emailed, and the gap between it and the real individual-effect average is the entire subject of this cookbook.

2 · From individuals to averages, and the enemy: confounding

Decisions run on **averages**: the **Average Treatment Effect** $ATE = \mathbb{E}[Y(1) - Y(0)]$. The whole difficulty collapses onto one word — **comparable**: to estimate $\mathbb{E}[Y(0)]$ for the treated we need a control group *like them in every way except the treatment*.

Confounding breaks comparability. If we email our *loyal* customers and compare to *casual* controls, loyal customers spend more anyway — “loyalty” drove both who got emailed and how much they spend. A **confounder** is a common cause of treatment and outcome; the naive “treated — control” difference then blames the treatment for a gap that was really about *who ended up in each group*.

3 · The gold standard, and life without it

Randomization (an A/B test) annihilates confounding by design: a coin flip makes treatment independent of every confounder, measured or not, so $ATE = \mathbb{E}[Y | T=1] - \mathbb{E}[Y | T=0]$ — a plain difference of means. We can *see* this: the same simulator run in **randomized** vs **observational** mode gives the same true effect, but only the randomized naive estimate is unbiased.

The data-generating model behind this demo — `dgp.uplift_customers`, the notebook-01 simulator (its full equations are in notebook 01, step 2). Each customer has features x (recency R , frequency F , monetary M , tenure, engagement E), a known baseline $\mu_0(x)$, and a planted effect $\tau(x)$, with spend

$$Y = \mu_0(x) + \tau(x)T + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 8^2).$$

The **only** thing that differs between the two bars is how T is assigned:

randomized: $T \sim \text{Bernoulli}(\frac{1}{2})$

observational: $T \sim \text{Bernoulli}\left(\sigma\left(1.8(E - 0.4) + 1.2 \frac{150-R}{150} + 0.15(F - 5)\right)\right)$,

where $\sigma(z) = 1/(1 + e^{-z})$ is the **logistic (sigmoid)** function — it maps any score to a probability between 0 and 1, so a higher score just means a higher chance of being emailed.

i.e. marketers historically emailed engaged, recently-active customers — the same characteristics that raise spend through $\mu_0(x)$. Identical true effect in both worlds; only the *assignment mechanism* changes, and that alone moves the naive estimate.

true ATE €5.6 * randomized naive €4.8 +/- 0.6 (SE) * observational naive €11.4 +/- 0.7 (SE)

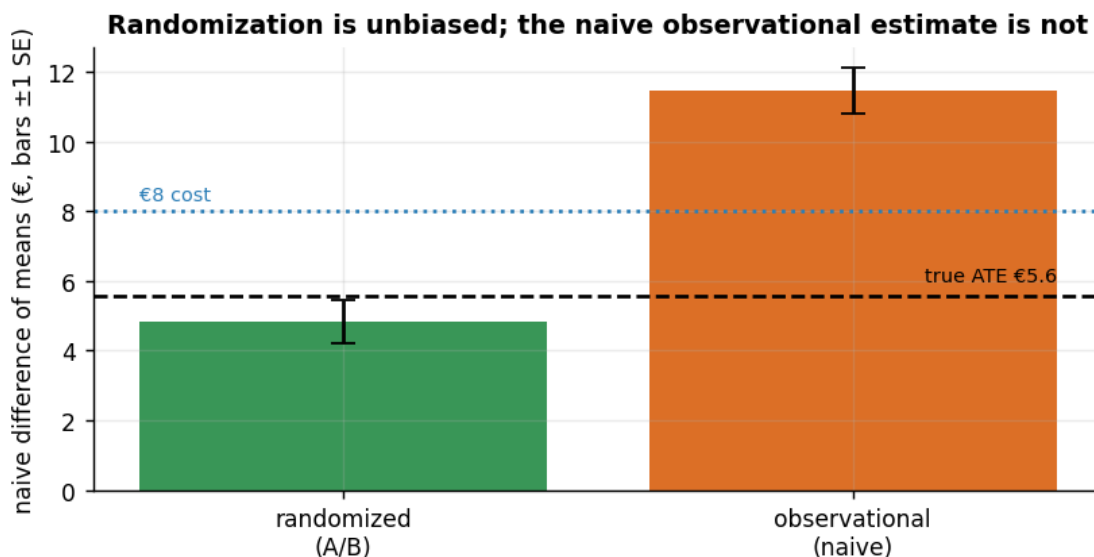
Same sampling noise (SE ~ €0.6 each), opposite reasons for missing the truth:

* randomized €4.8 is 1.2 SE from €5.6 - pure NOISE (unbiased; the gap shrinks like $1/\sqrt{n}$).

* observational €11.4 is 9.0 SE from €5.6 - that gap is BIAS (it does NOT shrink with n).

Decision at €8/contact: the TRUE effect €5.6 is BELOW €8, so emailing everyone does NOT pay.

But the confounded naive €11.4 is ABOVE €8 and would (wrongly) say 'email everyone' - the same identification error, now flipping a real go/no-go.



The anatomy of the naive difference — selection bias, named

The orange bar's failure has an exact algebraic name. Write the naive contrast with the two observed group means (always computable from a dashboard), use consistency ($Y = Y(1)$ when $T=1$, $Y = Y(0)$ when $T=0$), and add-and-subtract the counterfactual mean $\mathbb{E}[Y(0) | T=1]$ — what the emailed customers *would* have spent with no email:

$$\underbrace{\mathbb{E}[Y | T=1] - \mathbb{E}[Y | T=0]}_{\text{naive difference (observable)}} = \underbrace{\mathbb{E}[Y(1) - Y(0) | T=1]}_{\text{ATT: effect on the treated}} + \underbrace{\mathbb{E}[Y(0) | T=1] - \mathbb{E}[Y(0) | T=0]}_{\text{selection bias}}$$

Two new words, both load-bearing for the whole cookbook:

- **ATT** = $\mathbb{E}[Y(1) - Y(0) | T=1]$ — the *average treatment effect on the treated*: the effect among the customers who actually got the email. It differs from the ATE whenever treatment was *targeted* (here, marketers historically emailed exactly the customers the email works best on), and it is the **estimand** — the precise quantity a method targets, the four of which §5 lays out — that difference-in-differences (notebook 08) and the LaLonde data below actually identify.
- **Selection bias** = $\mathbb{E}[Y(0) | T=1] - \mathbb{E}[Y(0) | T=0]$ — how much more the emailed customers would have spent *even with no email*. A statement about **who** was chosen, not about what the email did.

Randomization makes T independent of $Y(0)$, so the selection term is **exactly zero** and $\text{ATT} = \text{ATE}$ — that is the entire one-line proof behind the green bar. The simulator lets us watch the identity balance, because it knows every customer's unobserved branch $Y(0) = Y - \tau T$:

naive difference	€ 11.45	
= ATT	€ 7.79	(effect ON THE EMAILED - targeting favoured)

high-effect customers)

+ selection bias € 3.66 (the emailed would have spent this much more ANYWAY)

check: ATT + bias = € 11.45 (the identity balances to machine precision, in-sample)

ATE (all customers) € 5.58 → the naive-ATE gap of €5.87 splits into
€3.66 of selection bias + €2.20 of targeting gap (ATT - ATE: past sends favoured high-tau customers).

Randomization zeroes the €3.66 selection term by construction ($T \perp\!\!\!\perp Y(0)$) and equates ATT with ATE -
that single fact is the whole content of the green bar above.

Without randomization we must **identify** the effect using assumptions:

1. **Unconfoundedness** $\{Y(0), Y(1)\} \perp T \mid X$ (the symbol $\perp$ reads “is independent of”) — within units alike on measured X , treatment is as-good-as-random. (*The big, untestable one.*)
2. **Positivity / overlap** $0 < e(x) < 1$, $e(x) = P(T=1 \mid X=x)$ the **propensity score** — every kind of unit could have gone either way.
3. **SUTVA** — no interference, one version of treatment.

Under these, the **adjustment (g-)formula** compares like-with-like *within* strata of X and averages back up: $ATE = \mathbb{E}_X[\mathbb{E}[Y \mid T=1, X] - \mathbb{E}[Y \mid T=0, X]]$.

The adjustment formula at work — stratify, compare like with like, average back up

Before the ladder automates it (§6), watch the g-formula work once **by hand**. For a discrete confounder the formula on means reads

$$\widehat{ATE}_{\text{strat}} = \sum_k \frac{n_k}{n} (\bar{Y}_{1k} - \bar{Y}_{0k}),$$

where k indexes strata of the confounder, n_k is the number of customers in stratum k (out of n total), and $\bar{Y}_{1k}, \bar{Y}_{0k}$ are the treated / control mean outcomes **within** stratum k : compare only like with like, then average the within-stratum contrasts, each weighted by its stratum’s population share. The demo data is `dgp.dag_control_demo` — the same dataset §6’s ladder and notebook 05 dissect (full equations in §6): one observed confounder, loyalty L , drives both who gets the email and spend, and the planted effect is exactly **€6**. We cut L into five quintile strata and run the formula:

stratum	loyalty range	n	% emailed	diff within (€)
Q1	[-3.5, -0.8]	600	25.83	6.08
Q2	[-0.8, -0.3]	600	41.83	5.64
Q3	[-0.3, 0.2]	600	48.67	6.32
Q4	[0.2, 0.8]	600	61.17	5.29
Q5	[0.8, 3.6]	600	76.33	6.74

pooled naive difference €9.26 (mixes the loyalty composition gap into

the 'effect')

stratified weighted average €6.02 (true planted effect €6)



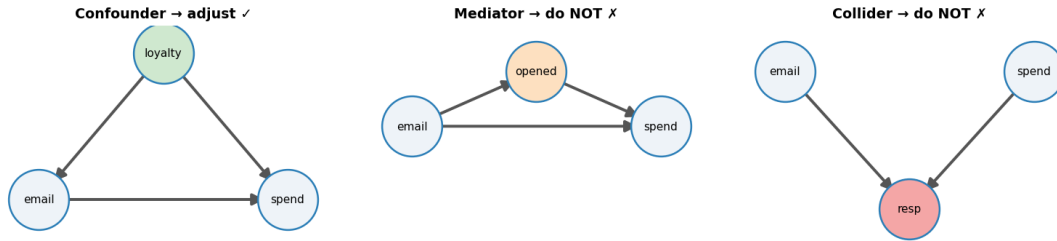
Read-out. *Within* like-loyalty customers the email is worth roughly the same $\approx$ €6 everywhere; the pooled $\approx$ €9 was never an email effect. The extra $\approx$ €3 is the composition gap the % **emailed** column makes visible — the share of emailed customers climbs steadily with loyalty, so the pooled contrast compares loyal emailed customers with casual controls. Every rung of §6’s ladder is a smarter way of doing exactly this stratification: **regression** replaces the five bins with a line, **IPW** re-weights each arm so the loyalty mix matches the population, **AIPW** does both at once. (Finer strata would shave the small residual within-stratum confounding — which is precisely why we hand the job to fitted models next.)

4 · Seeing vs doing (the do-operator), DAGs, colliders

Observing $X = x$ (units where X happened to be x , carrying all the reasons it got there) differs from **doing** $do(X = x)$ (reaching in and *setting* it). Causal effects are about *doing*. In notation: $P(Y | T=t)$ describes the units we *happened to observe* at $T=t$; $P(Y | do(T=t))$ describes what Y would look like if we *forced* everyone to t . Confounding is precisely their disagreement — in the stratification demo just above, “seeing” gave $\approx$ €9 while “doing” is worth the true €6. When the backdoor variables X are measured, the **backdoor adjustment formula** converts seeing into doing:

$$P(y | do(T=t)) = \sum_x P(y | T=t, X=x) P(X=x),$$

i.e. condition on $X=x$, then average over the *population* distribution of X — not over its distribution within an arm, which is what the naive contrast implicitly does. Taking means turns this into §3’s g-formula; the stratification table above **is** this equation executed by hand. A **DAG** draws each variable as a node and each direct cause as an arrow, and tells you exactly what to control for:



- **Backdoor path** (arrow *into* T, e.g. $T \leftarrow \text{loyalty} \rightarrow Y$): block it by conditioning — the minimal blocking set is the X in the adjustment formula.
- **Mediator** ($T \rightarrow M \rightarrow Y$): controlling for M removes part of the effect you want.
- **Collider** ($T \rightarrow C \leftarrow Y$): conditioning on C *manufactures* a spurious association. Notebook 05 is the whole “what to control for” story.

The third panel above draws exactly this: **response** is a *common effect* of email and spend (a collider), so conditioning on it opens a spurious email–spend path — which is why that panel is flagged “do NOT control.”

The collider trap, demonstrated with numbers (the third DAG panel, made real). The most common self-inflicted wound in marketing analytics is analysing only the customers who *responded* — opened, clicked, replied, converted. “Responded” is a **common effect** of the treatment and the outcome, and conditioning on a common effect *manufactures* association. Plant an email whose true effect is **exactly zero**:

$$T \sim \text{Bernoulli}(\tfrac{1}{2}), \quad Y = 20 + 0 \cdot T + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 5^2), \quad C = \mathbb{1}\{1.2T + 1.2\tilde{Y} + \eta > 0.8\}, \quad \eta \sim \mathcal{N}(0, 1),$$

where T is the (randomized!) email, Y is spend, $\tilde{Y}$ is spend standardized to unit variance, $\mathbb{1}\{\cdot\}$ is 1 when its bracketed condition holds and 0 otherwise, and C is “responded to the survey” — more likely if you were emailed *and* if you spend a lot. Because T is a coin flip, even the naive comparison is unbiased... until we filter on C :

email coefficient, all customers	: €+0.03	(true effect €0 - randomized, unbiased)
email coefficient, responders only (C=1)	: €-1.80	(44% of customers respond)

An email 'effect' out of thin air. Among responders, being emailed already explains the response, so the emailed responders needed LESS spend to get there - conditioning on the common effect induces a spurious negative email-spend link ('explaining away'). The same mechanics fire if you condition on opens, clicks or conversion: post-treatment variables stay OUT of the adjustment set. Notebook 05 dissects this fully - including a collider your drawn DAG cannot even see.

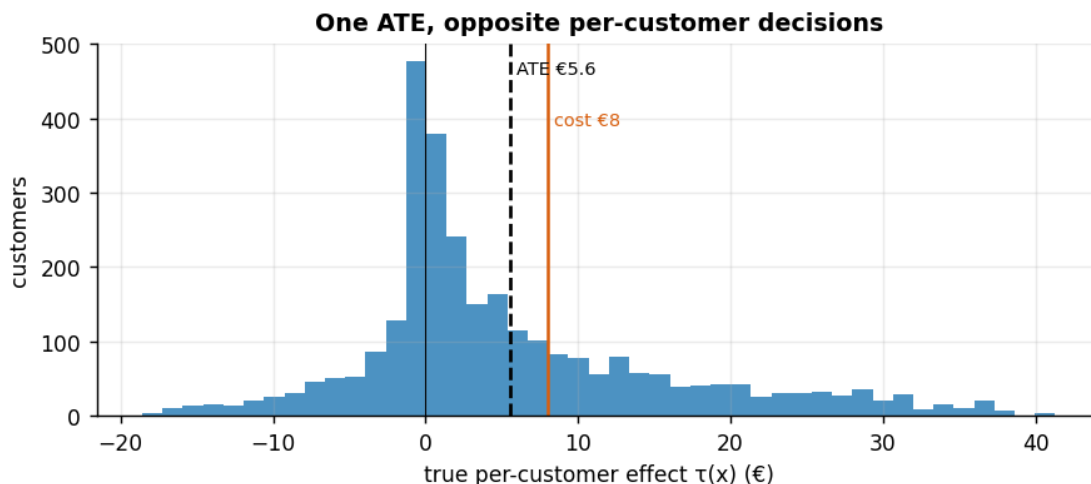
5 · Which average? — the estimand table (ATE, ATT, CATE, LATE)

“The effect” is never one number: before choosing a method you must name **whose** average you want — the **estimand**. The four that run this cookbook:

estimand	formula	plain meaning	who identifies it
ATE	$\mathbb{E}[Y(1) - Y(0)]$	effect on a <i>randomly chosen</i> customer — “does the program pay overall?”	A/B tests; the §6 ladder; program-level analogues in MMM / geo-lift / ITS (notebooks 06, 07, 10)
ATT	$\mathbb{E}[Y(1) - Y(0) \mid T=1]$	effect <i>on those actually treated</i> (defined in §3’s decomposition)	DiD (notebook 08), synthetic control (07) — and the LaLonde section below: its $\approx +\$1,800$ is an effect <i>on the trainees</i>
CATE	$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X=x]$	effect <i>for customers like x</i> — the targeting quantity	uplift / heterogeneity notebooks 01–04
LATE	$\mathbb{E}[Y(1) - Y(0) \mid \text{compliers}]$	effect on the units an <i>instrument</i> (a nudge that shifts exposure without otherwise touching the outcome) actually moves	IV (notebook 11)

They answer different business questions about the same campaign: ATE prices “*roll out to everyone?*”, ATT audits “*what did our past sends earn?*”, CATE decides “*who’s next?*”, and LATE is the honest scope when you could only *encourage* exposure, not assign it. §3’s decomposition already showed $\text{ATT} \neq \text{ATE}$ the moment treatment is targeted. The histogram below shows why even the right ATE can still be the wrong summary — targeting exists *because* $\tau(x)$ varies (if it were identical for everyone there would be nobody to select):

ATE €5.6 < €8 cost -> 'email no one'. Yet 29% of customers individually clear the €8 bar, and 30% have a NEGATIVE effect (the email backfires on them). One number, opposite per-customer calls.



The spread *is* the business case: at €8 a contact the ATE says stop, while roughly three customers in ten are individually profitable and a similar share would be actively harmed by the send. So whenever a stakeholder quotes “the” effect, the first question is *which estimand* (the table above) and the second is *how much it varies* — recovering the $\tau(x)$ that separates these groups is notebooks 01–04, and §9 below prices what that separation is worth in euros.

6 · The estimator ladder — four ways to recover one effect

This section is the *classical* read: **point estimates and confidence intervals** — **no likelihood, no priors, no sampler**. Everything below is estimated the way a competent analyst would estimate it without ever opening PyMC: a number, a standard error chosen to match the data (Welch for a difference of means, HC1 for a heteroskedastic cross-section, a percentile bootstrap where no closed form exists), and an interval built from it. In every *method* notebook (01–11) this move appears under its own heading, **Step 0 · the classical read**, before that notebook’s Bayesian section. In the foundations notebook it *is* §6 — and §7.1 below explains exactly what the Bayesian layer adds to it, and what it does not.

Where the machinery enters. On the **confounded** data (true effect €6), we climb a ladder of estimators, each leaning on different assumptions. On a **single** sample any estimate can land a little off — an honest 90% interval misses about 1 time in 10 — so the test of a *good* estimator is not “exact on this draw” but **unbiased**: centred on the truth *over repeated samples*, while the naive difference stays stubbornly biased. The right-hand panel below shows exactly that across many fresh samples.

1. **Naïve** difference of means — ignores X , biased up. (*Welch standard error: it does not assume the two arms have equal variances.*)
2. **Regression adjustment** — model $\mathbb{E}[Y | T, X]$, average the contrast (right if the outcome model is right). (*HC1 heteroskedasticity-robust standard error: spend is noisier for high-loyalty customers, and classical iid standard errors would understate that.*)
3. **IPW** — reweight by the inverse propensity so treated/control look alike (right if the propensity model is right). (*No closed-form standard error survives the estimated propensity, so:*

bootstrap.)

4. **AIPW (doubly-robust)** — combine both; consistent if *either* model is right, reported here with a **bootstrap confidence interval** — an interval built by re-drawing the sample *with replacement* many times and re-estimating, so the spread of those re-estimates measures the sampling uncertainty (AIPW is a frequentist estimator — this is *not* a Bayesian credible interval).

The standard-error column is not bookkeeping. An interval built on the wrong standard error is worse than no interval — it is a confident lie — so `cmp.classical` (the toolkit these fits use, `cl` in the code) makes the covariance choice explicit and refuses to default it silently.

The four estimators, written down. Let $\hat{m}(t, x) \approx \mathbb{E}[Y \mid T=t, X=x]$ be a fitted outcome model, $\hat{e}(x) \approx P(T=1 \mid X=x)$ a fitted propensity, and $\bar{Y}_1, \bar{Y}_0$ the raw arm means over the n customers. The four rungs are:

$$\begin{aligned}\hat{\tau}_{\text{naive}} &= \bar{Y}_1 - \bar{Y}_0, & \hat{\tau}_{\text{reg}} &= \frac{1}{n} \sum_{i=1}^n [\hat{m}(1, x_i) - \hat{m}(0, x_i)], & \hat{\tau}_{\text{IPW}} &= \frac{1}{n} \sum_{i=1}^n \left[\frac{T_i Y_i}{\hat{e}(x_i)} - \frac{(1 - T_i) Y_i}{1 - \hat{e}(x_i)} \right], \\ \hat{\tau}_{\text{AIPW}} &= \frac{1}{n} \sum_{i=1}^n \left[\hat{m}(1, x_i) - \hat{m}(0, x_i) + \frac{T_i(Y_i - \hat{m}(1, x_i))}{\hat{e}(x_i)} - \frac{(1 - T_i)(Y_i - \hat{m}(0, x_i))}{1 - \hat{e}(x_i)} \right].\end{aligned}$$

Why IPW works: weighting a treated unit by $1/\hat{e}(x_i)$ makes rarely-emailed kinds of customers count for more, so the re-weighted treated arm matches the **whole population’s** feature mix — the balance panel two cells down shows the loyalty imbalance — its **standardized mean difference (SMD)**, the gap in loyalty’s mean between the arms measured in pooled standard deviations — collapsing from far above the conventional **0.1** “balanced” threshold to ≈ 0 under exactly these weights. *Why double robustness works, in two lines:* AIPW is the regression answer plus a weighted-residual correction. If $\hat{m}$ is right, the residuals $Y - \hat{m}$ are mean-zero noise and the correction vanishes — you keep rung 2. If instead $\hat{e}$ is right, the correction is an unbiased IPW estimate of precisely the error a wrong $\hat{m}$ makes, and repairs it. Either model being correct suffices. In the code below, `reg` computes $\hat{\tau}_{\text{reg}}$ with a linear $\hat{m}$ (one `lstsq` line — for a linear model with a single T coefficient the average contrast *is* that coefficient), `ipw` computes $\hat{\tau}_{\text{IPW}}$ with a clipped logistic $\hat{e}$, and `est.aipw_ate` computes $\hat{\tau}_{\text{AIPW}}$ with **cross-fitting** (fit $\hat{m}$ and $\hat{e}$ on one half of the data and apply them to the other half, so no customer is scored by a model that saw it — this removes an overfitting bias in the correction term) plus a bootstrap interval.

The data-generating model for the ladder — `dgp.dag_control_demo`, the same dataset notebook 05 dissects (equations there too). One **observed confounder**, loyalty $L \sim \mathcal{N}(0, 1)$, and an independent unobserved trait $G \sim \mathcal{N}(0, 1)$ that adds outcome noise but never touches T (so it is *not* a confounder):

$$T = \text{email} \sim \text{Bernoulli}(\sigma(0.8L)), \quad Y = \text{spend} = 20 + 5L + 6T + 3G + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 4^2).$$

L raises both the emailing probability and spend — the open backdoor that rung 1 (naive) ignores and rungs 2–4 close, each in its own way: rung 2 models L in the outcome, rung 3 models L in the propensity, rung 4 does both. The true ATE the ladder must recover is the **6**.

The classical ladder on this sample - point estimate + 90% confidence interval, all in € (true €6):

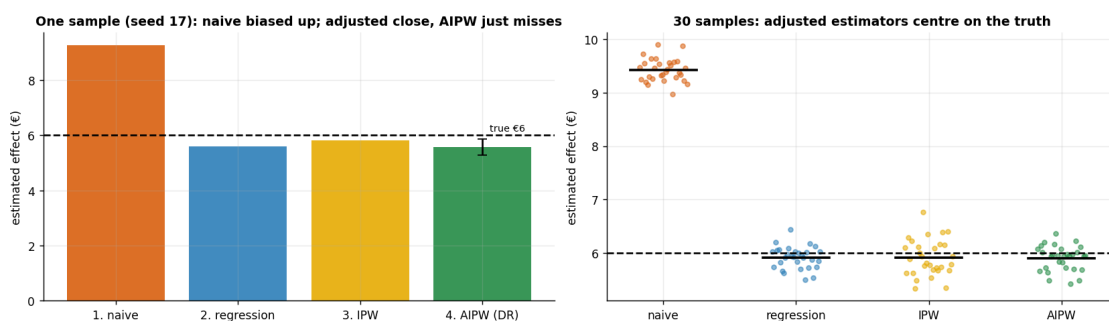
1. naive (difference in means)	9.26	[90% CI 8.84, 9.68]	*	SE 0.25	*
Welch (unequal variances)					
2. regression adjustment (OLS)	5.60	[90% CI 5.28, 5.92]	*	SE 0.19	*
HC1 heteroskedasticity-robust					
3. IPW	5.80	[90% CI 5.37, 6.21]	*	SE 0.26	*
percentile bootstrap (500 resamples)					
4. AIPW (doubly robust)	5.57	[90% CI 5.27, 5.86]	*	SE 0.20	*
cross-fitted influence function, bootstrap (300 resamples)					

What this 90% confidence interval does NOT say: that there is a 90% probability the true effect lies inside it. It is a property of the *procedure* - 90% of intervals built this way would cover the truth across repeated samples. This interval either contains the truth or it does not. To get a probability *about the effect itself* - the thing a go/no-go rule like $P(\text{lift} > \text{cost}) \geq 0.9$ actually needs - you need a posterior. That is what the Bayesian section adds.

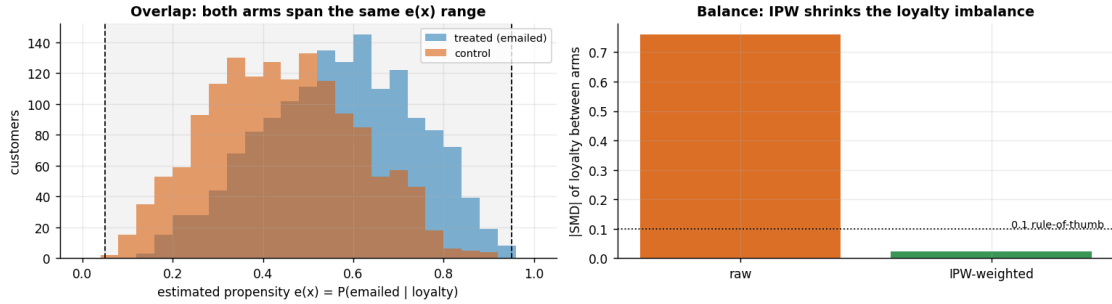
Across 30 fresh samples - mean estimate and bias vs the true €6:

naive	mean €9.42	bias +3.42	sd 0.21
regression	mean €5.91	bias -0.09	sd 0.20
IPW	mean €5.91	bias -0.09	sd 0.33
AIPW	mean €5.90	bias -0.10	sd 0.23

AIPW 90% interval covers the truth in 26/30 samples (87%); naive is the only estimator whose bias does not vanish.



overlap: e in $[0.06, 0.95]$; the clip binds on 0.0% of customers (rarely) - IPW shrinks the loyalty imbalance from SMD 0.76 to 0.02.



What adjustment cannot fix — the unmeasured backdoor

Everything so far is a success story, and that is a dangerous thing to teach: the ladder recovered €6 because the *one* confounder was **in the data**. Unconfoundedness (§3, assumption 1) is untestable, so the foundations question is *what happens when it fails*. We rerun the whole ladder on notebook 01’s customer simulator with a **hidden** trait U — think “a salesperson’s hunch” or an unlogged loyalty signal — wired into both assignment and outcome (the same two knobs notebook 01’s Depth B sweeps):

$$T \sim \text{Bernoulli}\left(\sigma(\text{observed-feature logit} + a_T U)\right),$$

$$Y = \mu_0(x) + \tau(x)T + b_Y U + \varepsilon, \quad U \sim \mathcal{N}(0, 1) \text{ unobserved},$$

with $a_T = 2.0$ (U shifts the emailing logit) and $b_Y = 3.0$ (U adds €3 of spend per SD). Every rung is fit on the five *observed* features only — exactly the information a real analyst would have:

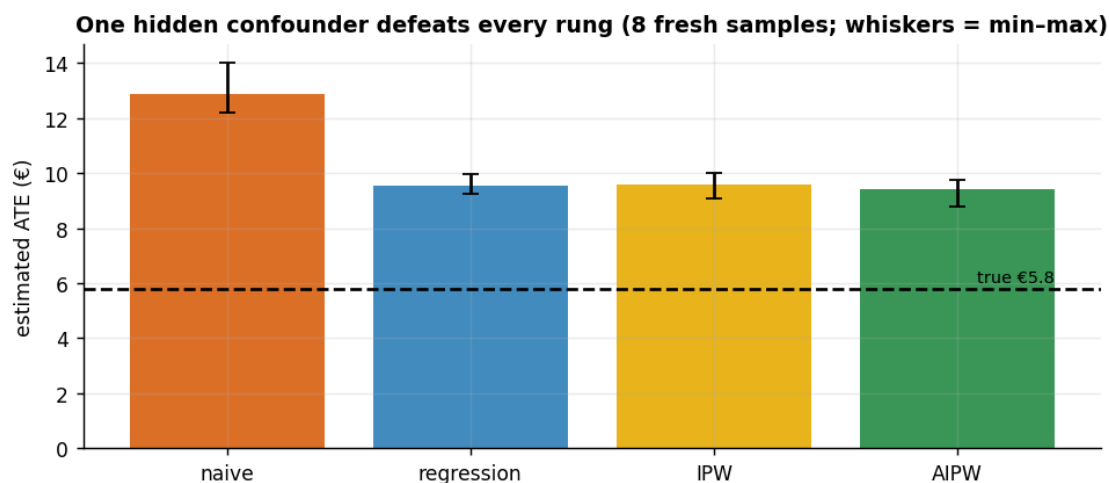
naive	mean €12.87	range [12.22, 13.99]	(true €5.79)
regression	mean € 9.55	range [9.23, 9.96]	(true €5.79)
IPW	mean € 9.59	range [9.09, 9.99]	(true €5.79)
AIPW	mean € 9.44	range [8.78, 9.74]	(true €5.79)

Every adjusted rung sits at least €3.6 above the truth: regression, weighting and double robustness fail TOGETHER, because all three adjust for the same measured features while the open backdoor runs through U , which none of them can see. No estimator upgrade fixes missing data.

Positivity failure, seen (10 seeds, true €6, confounder fully observed): under gentle selection IPW averages €6.12 (sd 0.26) and the clip never binds; under near-deterministic selection the clip binds on 44% of customers and IPW drifts to €7.51 (sd 0.42) — when some customers essentially can’t appear in both arms, no reweighting recovers a full-population

ATE. §6's overlap

histogram above is the diagnostic that catches this before you trust the number.



The boundary of estimation, drawn. Adjustment closes the backdoors you *measured*. A backdoor you didn't measure defeats every rung equally — and double robustness offers no protection, because it insures against a mis-specified *model of the observed features*, never against a missing feature. When positivity fails, the problem is worse still: the data simply contain no like-with-like comparisons to reweight. The honest responses are better data, better *design* (randomize — §3), or **sensitivity analysis**: asking how strong a hidden U would have to be to change the decision, which notebook 01's Depth B prices as a tipping point and an E-value. That boundary is why the discipline box below is the most important sentence in the cookbook:

The discipline: **Bayes/ML sharpens estimation, never identification.** A tight, beautiful posterior computed under a false unconfoundedness assumption is *confidently wrong*. Priors don't fix broken causal arguments — the graph does. Every notebook keeps identification (this ladder's assumptions) separate from estimation (which rung you climb).

7 · The Bayesian workflow, on one model — prior → likelihood → posterior → diagnostics

Everything above returned *points* (plus one bootstrap interval). From notebook 01 onwards every estimate in this cookbook is a **posterior distribution**, so the vocabulary lives here, on the smallest possible causal model: ladder **rung 2 made Bayesian**, on the same confounded data $\mathbf{d}$ ($n = 3,000$, true effect €6):

$$y_i \sim \mathcal{N}(\alpha + \tau T_i + \beta L_i, \sigma^2), \quad \alpha \sim \mathcal{N}(20, 10^2), \quad \tau \sim \mathcal{N}(0, 10^2), \quad \beta \sim \mathcal{N}(0, 10^2), \quad \sigma \sim \text{HalfNormal}(10).$$

Symbols: y_i is customer i 's spend (€), T_i the email indicator, L_i loyalty; α is the baseline spend of a loyalty-0 un-emailed customer, τ the email effect — **the estimand**, β the spend slope per SD

of loyalty (the term doing the causal adjustment), σ the residual spend noise. The first equation is the **likelihood** — exactly rung 2’s linear outcome model. The rest are **priors**, each defensible in one clause: all are *weakly informative on the € scale of the data* (α centred near typical spend of ~€20; β and σ wide enough for anything plausible), and $\tau \sim \mathcal{N}(0, 10^2)$ is *skeptical-but-open* — centred on “the email does nothing”, yet wide enough to learn a €30 effect if the data insist. The **posterior** $p(\tau \mid \text{data})$ is what MCMC samples below. **MCMC** (Markov-chain Monte Carlo) is the workhorse algorithm that, when the posterior has no closed form, draws a large sample of plausible τ values from it — each **draw** is one such value, and we run several independent **chains** (fresh restarts) so we can check they agree. The first **tune** draws are a warm-up the sampler discards while it calibrates itself; only the post-warm-up **draws** are kept — exactly what `dict(draws=400, tune=400, chains=2)` sets below. That code also prints a one-line sampler health check (R-hat, ESS, divergences); the box right after it defines those three numbers.

Credible vs confidence interval — say it right in the meeting. A 90% **credible interval** answers: *given the data and this model, there is a 90% probability the effect lies in this range* — the sentence managers actually want to hear. The confidence intervals of §6 answer a different question: *the procedure that built this interval captures the truth in 90% of repeated samples* — a statement about the procedure, not about this dataset. Different logic; numerically they nearly coincide when priors are weak and n is large (as here), and they part ways when data are thin — which is exactly when priors earn their keep (notebook 06’s MMM). §7.1 below makes that distinction exact and demonstrates it on the two fits this notebook has already run.

```
posterior for tau: mean €5.60 * 90% credible interval [€5.28, €5.93] (true €6)
```

```
AIPW (frequentist rung 4), same data: €5.57 * 90% bootstrap CI [€5.27, €5.86]
```

This sample’s 90% credible interval narrowly misses €6 — the SAME ~1-in-10 sampling event §6’s

AIPW interval showed on this seed: one dataset, one shared wobble; over fresh samples both centre on €6.

Bayesian and frequentist land together here: with weak priors and $n=3,000$ the likelihood dominates.

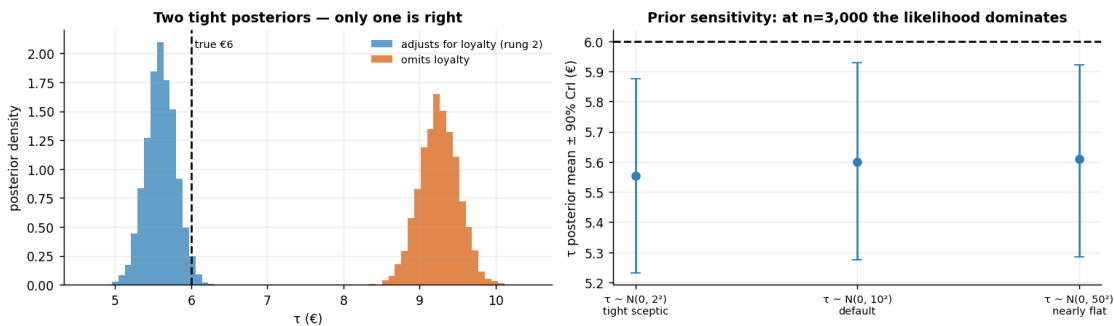
```
convergence: max r-hat 1.000 - min ESS 2513 - divergences 0 -> chains agree,
draws are plentiful, geometry is clean.
```

Did the sampler converge? — the three numbers to read before trusting any posterior. An MCMC posterior is only as good as the sampling that produced it, so every Bayesian fit in this cookbook prints the same one-line report (`est.convergence_report`). The vocabulary, in plain terms:

- **R-hat** — run several independent chains and compare where they think the posterior is. Agreement gives R-hat ≈ 1.00 ; above ~1.01, keep sampling or fix the model.
- **ESS** (*effective sample size*) — consecutive draws are autocorrelated, so 800 draws may carry the information of far fewer independent ones; ESS is that independent-equivalent count. You want it comfortably in the hundreds for stable interval endpoints.
- **Divergences** — NUTS (the No-U-Turn sampler, the default MCMC engine here) raising an alarm that it hit posterior geometry it couldn’t integrate; more than a handful means the posterior is suspect (raise `target_accept`, or reparameterize).

A four-parameter linear model is easy geometry, so this one passes without drama — the point is the **habit**. (Notebook 01’s BART models are sampled by particle Gibbs, where divergences don’t apply but R-hat/ESS still do; notebook 06’s MMM is hard enough geometry that FULL runs need `target_accept = 0.99`.)

same data, same priors, one dropped covariate: the tau posterior jumps €+3.66, to €9.26 +/- 0.25 — just as tight as the correct model’s (+/- 0.20) and centred on the naive rung’s answer, not on €6.
 prior sweep: the posterior mean moves <= €0.05 as the tau prior sd goes 2 -> 50 — with 3,000 customers the likelihood dominates. Priors matter when data are thin: notebook 06’s MMM fits saturation curves to 156 weeks, and there the priors carry real weight.



Both posteriors are beautiful. Only one is right. The no-loyalty model is not a worse *Bayesian* model — its chains converge, its R-hat is ≈ 1.00 , its interval is just as tight. It is a worse **causal** model: dropping L re-opened the backdoor, and the posterior obediently concentrated around the confounded answer. The prior didn’t fail — the **graph** did, and no prior on τ can encode “you forgot a confounder”. This is what *confidently wrong* looks like (the discipline box above, now demonstrated), and it is why every later notebook settles identification **before** admiring the posterior: the DAG decides *which variables enter the model*; Bayes only quantifies uncertainty about their coefficients. The prior-sensitivity panel adds the complementary calibration: with $n = 3,000$ the prior is nearly irrelevant — so when a stakeholder worries that “the priors drive the answer”, the sweep is the reply; and when data are thin (notebook 06), the same sweep is how you find out that they do.

7.1 · Point estimate vs posterior — the distinction the rest of this cookbook rests on

§6 produced a number and an interval. §7 produced a distribution. On the page they look almost identical — the same effect, roughly the same range — and that resemblance conceals the single idea every later notebook leans on. So state it once, exactly, and re-use it.

A confidence interval is a property of the procedure. Treat the effect τ as a fixed, unknown constant; the *data* are what is random. An interval estimator $[L(D), U(D)]$ has 90% coverage when

$$\Pr_{D \sim P_\tau} (L(D) \leq \tau \leq U(D)) = 0.90 \quad \text{for every possible } \tau.$$

Read where the probability sits: on D , the dataset that could have come out otherwise — **not** on τ , which is not a random variable at all in this calculus. Once the data are in hand, the printed interval either contains τ or it does not. There is no “90% chance” left in it, exactly as a coin already flipped and covered by your hand is not “50% heads” — you are simply ignorant of a settled fact. Ninety percent describes the **factory**, not this one interval off the line. And §6 audits that factory in the only way it can be audited: rebuild the interval on fresh samples and count how often it catches the planted €6 (the coverage line printed under the ladder).

A credible interval is a probability statement about the effect. Put a distribution on τ itself — a prior — and let the likelihood update it into a posterior $p(\tau \mid D)$. A 90% credible interval is any $[a, b]$ with

$$\Pr(a \leq \tau \leq b \mid D, \text{ model, prior}) = 0.90.$$

Now the probability *is* over τ , conditional on the data you actually have. This is the sentence everyone believes they are saying when they read a confidence interval aloud in a meeting. It is a legitimate sentence — but only after paying its price of admission: a model and a prior, both of which must then be defended (§7’s prior sweep, notebook 05’s DAG).

Why this is commercial, not philosophical. Every decision in this cookbook has the shape

$$\Pr(\text{lift} > \text{cost} \mid \text{data}) \geq 0.9 \implies \text{ship it},$$

and that expression is **natively a posterior quantity**: it integrates a distribution over the parameter above a threshold. A confidence interval has no distribution over the parameter to integrate — there is nothing there to take the mass of. The classical arm *can* manufacture a look-alike: bootstrap the estimator, then count what share of the re-estimates clear the cost. But the object it produces is the sampling distribution of the **estimator** $\hat{\tau}$ — how the *estimate* would jump around across resampled datasets — and not a distribution of belief over the **parameter** τ . The two land on nearly the same number in the friendly case (large n , weak priors, near-symmetric likelihood — the case below, which is precisely why the confusion is so durable and so hard to kill). They separate exactly where the money is: thin data, hierarchical shrinkage across segments, a prior doing real work, and any nonlinear function of the parameters — ROAS, a saturation curve, the value of a targeting policy — where uncertainty must be *propagated*, not just reported. Reading a bootstrap as a belief is borrowing the Bayesian sentence without paying for the model. (If you *are* willing to pay — flat prior, regular likelihood — the two coincide by a theorem, not by luck; but then you have simply become a Bayesian.)

The demo below is deliberately anticlimactic. Both arms have already been fitted, on the same 3,000 customers, targeting the same τ : the classical OLS rung 2 of §6 and its Bayesian twin of §7 — the *same* linear specification, with priors bolted on. We put them side by side, and then ask the one question only one of them can answer.

Same estimand tau (true €6), same data (n=3,000):

	arm	estimate €	lo €	hi €	width €	interval
classical * OLS rung 2 (HC1)		5.60	5.28	5.92	0.64	90% CONFIDENCE
classical * AIPW (bootstrap)		5.57	5.27	5.86	0.60	90% CONFIDENCE
Bayesian * rung 2 + priors		5.60	5.28	5.93	0.65	90% CREDIBLE

rung 2, both ways: the point estimates differ by €0.002 (0.3% of the confidence interval's width), and the two intervals agree to within a couple of cents. Bayes did NOT change the answer.

$P(\tau > €5.25 \mid \text{data}, \text{model}) = 0.96$ <- posterior mass above the cost. A probability ABOUT THE EFFECT:

the number the rule 'ship if $P(\text{lift} > \text{cost}) \geq 0.9$ ' literally asks for. Verdict at the 0.9 bar: SHIP.

Bootstrap look-alike: 0.96 of the 2,000 re-estimates of τ_{hat} sit above €5.25.

That is NOT $P(\tau > \text{cost})$. It describes how the ESTIMATOR would scatter across resampled datasets,

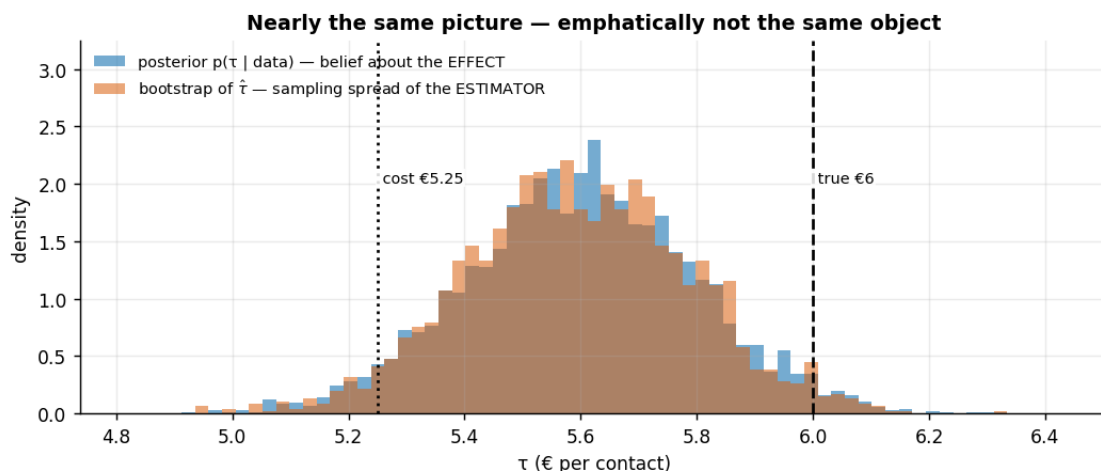
not how plausible each value of τ is. It is numerically close here - weak priors, $n=3,000$, a

symmetric likelihood - and that near-coincidence is exactly why the mistake survives in industry.

And no confidence interval can be read that way at all: §6's ladder can only tell you that intervals

built that way covered the planted truth in 26/30 of the fresh samples it drew (87%) -

a fact about the procedure. Ask it for $P(\text{lift} > \text{cost})$ and it has no distribution over τ to integrate.



What the demo shows, in three beats.

1. **They agree.** The classical OLS rung 2 and its Bayesian twin land on the same effect, with intervals that overlap almost perfectly (printed above — a difference of a few cents on a €6 effect). This is the expected result, not a disappointment, and it is worth saying plainly to anyone who suspects Bayes of smuggling in the answer it wants: with $n = 3,000$ and weakly-informative priors the likelihood dominates, and both arms are just reading the same data honestly.
2. **Only one of them answers the business question.** $P(\tau > \text{cost} \mid \text{data})$ is a single line of arithmetic over the posterior draws — count the fraction above the threshold — and it is *the* input to the $P(\text{lift} > \text{cost}) \geq 0.9$ rule that every €-denominated decision in this cookbook uses. The classical arm has a point estimate and a coverage guarantee, and neither can be turned into that probability. The bootstrap number printed next to it is a different object wearing the same clothes: the scatter of the *estimator*, not the plausibility of the *effect*.
3. **What Bayes did *not* buy here.** Nothing in precision: the credible interval is no tighter than the HC1 confidence interval, and §7's prior sweep moved the posterior mean by cents as the prior went from sceptical to nearly flat. On easy data like this, the Bayesian layer buys **interpretability and a decision rule**, and that is all — say so. Where it buys more is where the data stop being easy: partial pooling across thin segments (notebook 02), priors carrying real weight on 156 weeks of MMM (06), uncertainty propagated through a saturation curve into ROAS (06) or through a policy into euros (01), and coherent sensitivity analysis over an assumption you cannot test (01's Depth B).

How to read the rest of this cookbook. Every method notebook (01–11) opens its estimation work with **Step 0 · the classical read:** the same estimand, the same identification assumptions, the simplest *correct* estimator — a 2×2 difference-in-differences, 2SLS, a local-linear regression discontinuity, an SLSQP synthetic control — fitted with the standard errors that data structure demands (cluster on panels, HAC on time series, HC1 on cross-sections), and *then* the Bayesian model. Two reasons, both on display above. First, it lets you meet the **causal idea** — what identifies the effect — before meeting the **Bayesian machinery**, instead of both at once; the causal content lives in the identification argument, which is the same in either arm. Second, it keeps the comparison honest: when the two arms agree, the Bayesian layer is not doing the work, and you should know that. Each notebook ends its Step 0 on the same guardrail sentence printed under §6's ladder (`cmp.classical's Classical.cannot_say()`), and closes the loop with its own point-estimate-vs-posterior read-out.

When one of those read-outs feels slippery — and eventually one will — come back to **§7.1 · Point estimate vs posterior**. It is the definition the whole cookbook cross-references.

8 · The same lesson on real, famous data (LaLonde 1986)

The ladder above ran on our simulator. Here is the identical confounding story on the most famous real dataset in causal inference — **you don't provide anything; it's fetched from a public URL:**

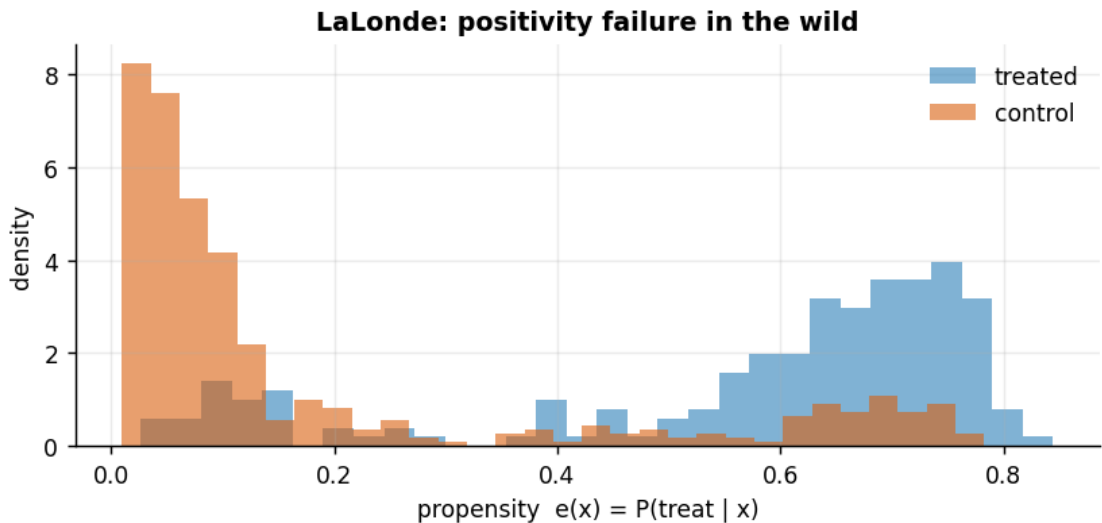
LaLonde / NSW — a 1970s job-training program. Because it included a **randomized** arm, we know the *true* effect of training on 1978 earnings: about **+\$1,800**. But if we throw away the randomized controls and instead compare the trained workers to a *non-experimental* comparison group (people pulled from national surveys), a naive difference is **negative** — it looks like training *hurt* earnings, because the trained group was far more disadvantaged to begin with (the confounding). Adjusting for the measured

covariates pulls the estimate back toward the true +\$1,800. This single dataset is why the field takes confounding seriously.

Gated on CMP_REAL=1 (keeps the offline test suite deterministic).

LaLonde (observational comparison group), effect of training on 1978 earnings:
naive difference of means : -\$635 (biased - trained group started far poorer)
covariate-adjusted : \$1,548 (moves toward the ~\$1,800 experimental truth)

overlap: 60% of the survey controls sit at $e(x) < 0.10$ vs 5% of trainees - regression is extrapolating across thin support for most of the control pool, which is why the adjusted estimate above still sits short of the ~\$1,800 experimental truth, and why the field moved to propensity trimming/matching (Dehejia-Wahba 1999) and the doubly-robust AIPW rung.



The payoff (what CMP_REAL=1 prints). Fetching the real LaLonde/NSW data ($n = 614; 185$ trained) and running the *same two rungs* as our simulator gives:

estimator	effect on 1978 earnings	reading
naive difference of means	-\$635	training looks <i>harmful</i> — the comparison group started far richer
covariate-adjusted	+\$1,548	pulled back toward the $\approx$ +\$1,800 experimental truth

Two more lessons the gated run adds. (i) **Positivity failure in the wild:** the propensity histogram the gated cell draws shows roughly **60% of the survey controls at $e(x) < 0.1$** while almost all trainees sit higher — LaLonde is the canonical overlap-failure dataset, the real-data twin of §6’s overlap panel. (ii) **The honest residual:** covariate adjustment on that poor overlap lands at **+\$1,548** against the $\approx$ **+\$1,800** experimental benchmark — right sign, right order, still ~\$250 short. Closing that residual on bad overlap is what propensity trimming/matching (Dehejia–Wahba 1999) and doubly-robust estimators (§6’s AIPW rung) were invented for.

This is an even starker flip than the simulator’s go/no-go — here the *sign* itself **reverses** once you adjust, driven by the same confounding, on the dataset that made the field take confounding seriously. (The committed offline run above shows only the skip message so the test suite stays deterministic; these numbers were produced by the gated cell with `CMP_REAL=1`.)

9 · Price the identification error — the same mistake, in euros

Statistical bias becomes a business problem the moment someone acts on it. §3’s confounded read-out (naive $\approx$ €11 per contact vs €8 cost) says *email everyone*; the true ATE ($\approx$ €5.6) says *a blanket send loses money at this cost*. The simulator knows every customer’s true τ_i , so we can price what each recommendation would actually have earned on this 3,000-customer base — plus the per-customer **oracle** (email exactly those with $\tau_i > \text{€8}$), which no ATE-level rule can reach:

policy	decided by	realised €
email everyone	naive €11.4 > €8 (confounded)	-7255
email no one	true ATE €5.6 < €8 (right on average)	0
oracle: email $\tau_i > \text{cost}$	per-customer truth (nb 01's target)	9481

Acting on the confounded number burns €7,255 on this base - identification error is a P&L

line, not a statistics footnote. And the oracle row shows €9,481 sitting in per-customer

targeting that NO average-level decision can touch: that gap is notebook 01's entire subject.

The decision memo (what you tell the CMO). The dashboard says the email pays for itself; it doesn't. Most of the gap is *who we emailed*, not what the email did — our historical targeting picked loyal, engaged customers who would have spent more anyway (**selection bias**), and on whom the email also works better than average (**ATT above ATE**); §3’s decomposition splits the gap exactly, and the table above prices acting on it at roughly €7,000 burned per 3,000 contacts. What to do instead, in order: **randomize** when you can — a holdout kills selection bias by design (§3’s green bar); when you can’t, **adjust** for the confounders you measured (§6’s ladder) and **stress** the ones you didn’t (§6’s hidden-*U* demo; notebook 01’s Depth B prices that stress as a tipping point and an E-value). And note where the real money sits: not in the average at all, but in per-customer targeting (the oracle row) — which is exactly notebook 01’s opening question.

10 · The cookbook map — every notebook, its estimand, and the assumption that carries it

Skim this once now and again whenever you open a new notebook — it is the intro’s promise (“every later notebook reads as a *named tool*”) in one table, and a decent revision slide. Every one of them opens its estimation work with **Step 0 · the classical read** — the same estimand and assumptions, the simplest correct estimator, a point estimate and a confidence interval — before it fits anything Bayesian; the reason is §7.1 · Point estimate vs posterior, the passage those notebooks cross-reference by name. Every “breaks when...” entry is that notebook’s falsification section: the foundations habit — name the assumption, then attack it — scaled up.

notebook	method	estimand	key identification assumption	breaks when...
01 uplift targeting	BCF / meta-learners	CATE	unconfoundedness + overlap	a hidden driver steered past sends (Depth B prices it)
02 segment effects	hierarchical Bayes	segment CATEs	as 01, within segments	the segment label proxies for the confounder itself
03 price elasticity	hierarchical log-log	elasticity (continuous T)	price moves $\perp$ demand shocks, given controls	simultaneity — managers cut price when demand is soft
04 funnel mediation	SCM / natural effects	direct + indirect effects	sequential ignorability	an unmeasured mediator–outcome confounder
05 what to control for	DAGs + adjustment sets	(whichever you target)	your DAG is right	you conditioned on a mediator/collider — or can’t see one
06 incrementality MMM	Bayesian MMM	channel contributions / ROAS (return on ad spend)	seasonality & spend confounding modelled	collinear channels, thin data (priors carry real weight)
07 / 07b geo lift	synthetic control	ATT (treated geo)	donors reconstruct the counterfactual (pre-fit transfers)	interference, shocked donors, short pre-periods
08 loyalty rollout	DiD	ATT	parallel trends	treated stores were trending differently anyway
09 threshold perk	RDD	local ATE at the cutoff	continuity — no sorting at the threshold	customers game the threshold (McCrary check)

notebook	method	estimand	key identification assumption	breaks when...
10 site redesign	ITS	post-period effect on the treated series	the pre-trend model extrapolates cleanly	anything else launched the same week
11 / 11b ad exposure	IV	LATE (compliers)	exclusion + a strong first stage	weak instrument, or one that touches the outcome directly

The ten words to walk in with

Treatment · **counterfactual** / **potential outcome** (the branch you didn't observe) · **confounder** (common cause of treatment and outcome) · **selection bias** (the who-was-chosen part of the naive gap — §3's decomposition) · **randomization** (kills confounding by design) · **identification** (the assumptions that make an effect recoverable) · **do-operator** (forcing vs observing) · **backdoor** / **collider** (what to control for, and what never to) · **ATE** / **ATT** / **CATE** (whose average — §5's estimand table) · **posterior & credible interval** (given data and model, how sure we are — §7's workflow, guarded by R-hat/ESS) — as opposed to a **confidence interval** (a property of the *procedure*, not a probability about the effect: §7.1, the section every later notebook's Step 0 points back to).

Next: [01 · uplift targeting](#) (Anchor A, HTE) and [07 · geo-lift synthetic control](#) (Anchor B, program evaluation) — then the rest of the cookbook, with §10's map as the index.